Discussion and Analysis

H_2 and HeH^+ with STO-nG Basis Sets ($n = 2$–6)

1 Introduction

Dissociation curves for H_2 and HeH^+ using the Restricted Hartree–Fock (RHF) method with minimal STO-nG basis sets were computed to understand how the energy changes with basis size. Both molecules are two-electron closed-shell systems, which makes them the simplest possible test cases for RHF.

The calculations were done in Python using the same analytical integral machinery described in the theory section: overlap, kinetic, nuclear attraction, and electron repulsion integrals are all computed in closed form using the Gaussian product theorem and the Boys function. Bond lengths were scanned in steps of 0.01 Å and the total energy $E_{\text{total}} = E_{\text{elec}} + V_{\text{nuc}}$ was recorded at each point.

2 Basis Sets

The STO-nG basis approximates the Slater 1s orbital as a sum of n Gaussian primitives. One contracted function is placed on each atom, giving a minimal basis of dimension 2 for both molecules. The normalisation constant for each primitive is $N(\alpha) = (2\alpha/\pi)^{3/4}$, and the contraction coefficients and exponents are taken from the Basis Set Exchange.

As the exponents for He are roughly four times larger than those for H at the same n, reflecting the tighter 1s orbital of helium ($Z = 2$). The minimal basis converges more slowly for He than for H. The computational complexity also increases with n.

3 Results

3.1 Equilibrium geometries and binding energies

The following table summarises the equilibrium bond length r_e and binding energy D_e extracted from the potential energy curves. The binding energy is defined as

$$D_e = E_{\text{min}} - E_{\text{dissociation}}, \tag{1}$$

Molecule	Basis Set	r_e (Å)	E_{min} (Ha)	E_{dis} (Ha)	D_e (Ha)
H2	STO-2G	0.72	-1.0938	-0.6300	0.4639
H2	STO-3G	0.71	-1.1175	-0.6560	0.4615
H2	STO-4G	0.71	-1.1240	-0.6630	0.4609
H2	STO-5G	0.71	-1.1257	-0.6650	0.4606
H2	STO-6G	0.71	-1.1262	-0.6657	0.4606
HeH+	STO-2G	0.91	-2.7653	-2.7022	0.0631
HeH+	STO-3G	0.93	-2.8544	-2.8078	0.0465
HeH+	STO-4G	0.94	-2.8782	-2.8358	0.0425
HeH+	STO-5G	0.94	-2.8851	-2.8438	0.0413
HeH+	STO-6G	0.94	-2.8873	-2.8463	0.0409

For H_2, the equilibrium bond length settles at 0.710 Å from STO-3G onward, about 30 mÅ shorter than experiment. This is typical for RHF, which tends to overbind at short range because the mean-field approximation overestimates the exchange stabilisation. The minimum energy decreases steadily with n, consistent with the variational principle.

For HeH^+ the STO-2G result stands out. This is because the He 1s orbital is more compact and harder to approximate with only two Gaussians.

3.2 Basis set convergence

There is incremental improvement in $E_{\min}$ at each increasing n. The pattern is the same for both molecules: the biggest gain is always from STO-2G to STO-3G, and the improvements shrink at each step. By STO-6G, the energy is essentially saturated within the single-contracted-s approximation; the remaining gap to the HF limit is not a basis-set issue but a fundamental limitation of representing a Slater orbital with Gaussians at a single centre.

The HeH^+ improvements are about four times larger at every step, which is consistent with the harder-to-represent He orbital discussed above.

3.3 The dissociation curve

The full dissociation curves are shown in Figure 1.

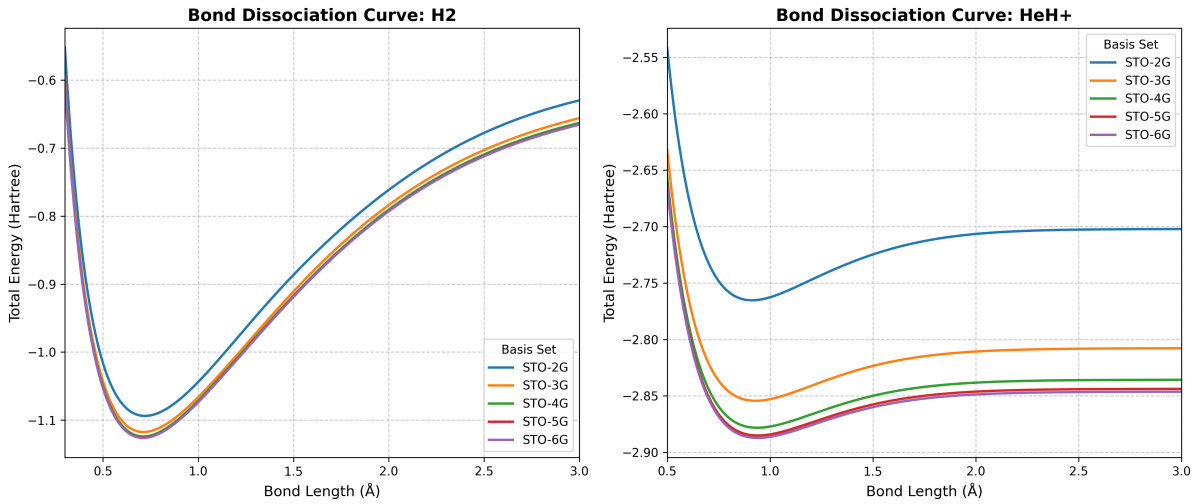

Figure 1: Bond dissociation curves for H_2 (left) and HeH^+ (right) across STO-2G through STO-6G. The dashed lines in the original scan mark the correct dissociation limits at each basis level.

Both curves show the correct qualitative shape: a minimum at the equilibrium geometry, a steeply repulsive wall at short range due to nuclear repulsion, and a plateau at large R. However, the plateau values tell very different stories for the two molecules.

For HeH^+, the curves flatten out close to $E(\mathrm{He})$ at large R, which is the correct result. This works because HeH^+ dissociates into He (neutral, 2 electrons) and H^+ (no electrons), and a single closed-shell determinant describes this correctly all the way to dissociation.

For H_2, the situation is quite different. At $R = 3.0$ Å the RHF energy lies well above $2E(\mathrm{H})$. This is not a basis-set problem, since the error barely changes across all five basis sets.

The reason for this failure is that RHF forces the wavefunction to be a single closed-shell Slater determinant throughout. At large R, the σ_g^2 configuration can be expanded as

$$|\sigma_g \bar{\sigma}_g\rangle \propto \underbrace{(\phi_A\phi_B + \phi_B\phi_A)}_{\text{covalent}} + \underbrace{(\phi_A\phi_A + \phi_B\phi_B)}_{\text{ionic}}, \tag{2}$$

where covalent (one electron on each atom) and ionic (both electrons on one atom) contributions have equal weight. At large separation, the ionic configurations H^+H^- carry a large Coulomb penalty, but RHF cannot reduce their weight because it is constrained to a single determinant. The error is intrinsic to the method and can only be fixed with a multi-reference treatment, such as CASSCF or GVB.

4 Conclusion

The main results are as follows:
The basis set converges quickly and variationally for both molecules, with most of the gain coming from the 2G→3G step. Convergence is slower for HeH^+ due to the compact He orbital. The equilibrium geometry and minimum energy are well described by STO-6G within the limitations of the minimal basis. The fundamental failure of RHF is not being able to to correctly dissociate H_2: a basis-independent error arises from the forced equal mixing of covalent and ionic configurations in the single-determinant wavefunction. HeH^+ dissociates correctly because it separates into a closed-shell atom and a bare proton, a situation the single determinant handles without difficulty.